

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iv)	Potential equation used (e.g. 630 000, -282 000) (1) Potentials added and subtracted (696 000) (1) WD = $q\Delta V$ used (± 0.0264) (1) Energy (not potential) equated to $\frac{1}{2}mv^2$ (1) Final answer (8.3 m s^{-1}) (1) Alternative: Potential equation used (-0.02394, +0.0107) (1) PEs added and subtracted (1) Value correct (± 0.0264) (1) Equated to $\frac{1}{2}mv^2$ (1) Final answer (8.3 m s^{-1}) (1) Award a maximum of 3 marks for: Any attempt at $a = \frac{F}{m}$ (1) expect $a = 73.3$ Any attempt at use of equations of motion (1) i.e. $v^2 = u^2 + 2ax$ Answer of $9.4 \text{ [m s}^{-1}\text{]}$ (1) Answers that score 4 marks: 5.86 $\text{[m s}^{-1}\text{]}$, 2 charges instead of 4 Power of 10 – cm instead of m should be 83 $\text{[m s}^{-1}\text{]}$, same applies to mC instead of nC – out by a factor of 1000. Unfortunately, nC instead of μC will be out by a factor of $\sqrt{1000}$ = 31.6 $\text{[m s}^{-1}\text{]}$ 7.46 $\text{[m s}^{-1}\text{]}$ and 11.15 $\text{[m s}^{-1}\text{]}$	1 1	1 1 1		5	3	
			Question 4 total	6	9	3	18	7	0